STAT 5700 — Quiz 6
Date: April 16, 2026

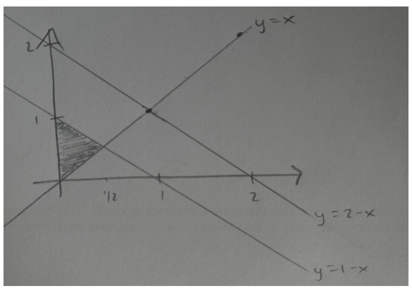
SOLUTIONS

FOR ALL 3 PROBLEMS: Let X and Y be two continuous random variables with join pdf

$$f(x, y) = 3xy,\ 0 < x < y,\ x + y \leq 2$$

Problem 1 (3pts) Find P(X + Y < 1)

SOLUTION



$$\begin{aligned}
P(X+Y<1) &= \int_0^{1/2} \int_x^{1-x} 3xy\,dy\,dx \\
&= \int_0^{1/2} 3x \left[\frac{y^2}{2}\right]_{y=x}^{1-x} dx \\
&= \int_0^{1/2} \frac{3x}{2}\left((1-x)^2 - x^2\right) dx \\
&= \int_0^{1/2} \frac{3x}{2}(1-2x)\,dx \\
&= \int_0^{1/2} \left(\frac{3}{2}x - 3x^2\right) dx \\
&= \left[\frac{3}{4}x^2 - x^3\right]_0^{1/2} \\
&= \frac{3}{16} - \frac{1}{8} \\
&= \frac{1}{16}.
\end{aligned}$$

Problem 2 (1pt): Find an expression for the marginal distribution of X. You can leave your answer written as an integral, but make clear what function you are integrating and what the appropriate bounds are.

$$f_X(x) = \int_x^{2-x} 3xy\,dy$$

Problem 3 (1pt): Find an expression for $f(y|x = \frac{1}{2})$

$$f(y \mid x = \tfrac{1}{2}) = \frac{f(0.5,\, y)}{\int_{0.5}^{1.5} 3(0.5)y\,dy} = \frac{3(0.5)y}{\int_{0.5}^{1.5} 3(0.5)y\,dy} = y, \quad 0.5 < y < 1.5$$

$$f_X\big(\tfrac{1}{2}\big) = \int_{y_{\min}}^{y_{\max}} f_{X,Y}\big(\tfrac{1}{2},\, t\big)\ dt$$

$$f(y \mid x = \tfrac{1}{2}) = \frac{f_{X,Y}\big(\tfrac{1}{2},\, y\big)}{\displaystyle\int_{y_{\min}}^{y_{\max}} f_{X,Y}\big(\tfrac{1}{2},\, t\big)\ dt}$$